



Chapter S4G: Network Security

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4.3.1 Understand how malware (e.g. worms and viruses) and denial of service (DOS) attacks can compromise computer systems.

4.3.2 Understand how firewall (filtering function), intrusion detection system (IDS) and intrusion prevention system (IPS) can be used to restrict network access, and their limitations.

4.3.3 Understand how encryption, digital signature, and authentication can ensure security of network applications.



1. Introduction to Network Security

Network security refers to measures used to protect computer networks and data from unauthorised access, misuse, modification, or disruption.

The main goals of network security are:

Confidentiality	Prevent unauthorised access to data
Integrity	Prevent unauthorised modifications of data
Availability	Ensure systems and services remain accessible

Network security uses both hardware and software technologies such as firewalls, antivirus software, encryption, and authentication systems.

2. Malware and Viruses

Malware (malicious software) refers to software designed to damage, disrupt, or gain unauthorised access to a computer system. Examples of malware include viruses, worms, Trojan horses, ransomware, and spyware.

Malware Comparison

Malware Type	Self-Replicates	Needs Host File	Requires User Action
Virus	Yes	Yes	Usually yes
Worm	Yes	No	No
Trojan Horse	No	No	Usually yes

Virus

A virus is malware that attaches itself to legitimate files or programs. It spreads when infected files are executed, usually requiring user action such as opening an infected attachment. Viruses may corrupt files, delete data, or execute unauthorised actions.

Worm

A worm is self-replicating malware that spreads across networks without needing a host file or user action.

Trojan Horse

A Trojan horse is malware disguised as legitimate software to trick users into installing it. Once installed, it performs malicious actions but does not self-replicate.

Ransomware

Ransomware encrypts files or locks a computer system and demands payment to restore access.

Spyware



Spyware secretly collects information such as passwords, browsing activity, or personal data without the user's knowledge.

Anti-virus software

Anti-virus software detects and removes malware by scanning files and memory for known malware signatures or suspicious behaviour. It may also provide real-time protection by scanning files as they are opened or downloaded.

3. Denial of Service

A Denial of Service (DoS) attack occurs when a system, server, or network is overwhelmed with excessive requests or traffic, causing legitimate users to be unable to access the service.

DoS attacks may:

- exhaust system resources
- overload network bandwidth
- crash servers
- disrupt normal operations

Services affected may include:

- web servers
- email servers
- file-sharing services
- online applications

Distributed Denial of Service (DDoS)

A Distributed Denial of Service (DDoS) attack is a DoS attack launched from many compromised computers or devices simultaneously, often forming a botnet. The large volume of traffic makes the target service unavailable to legitimate users.



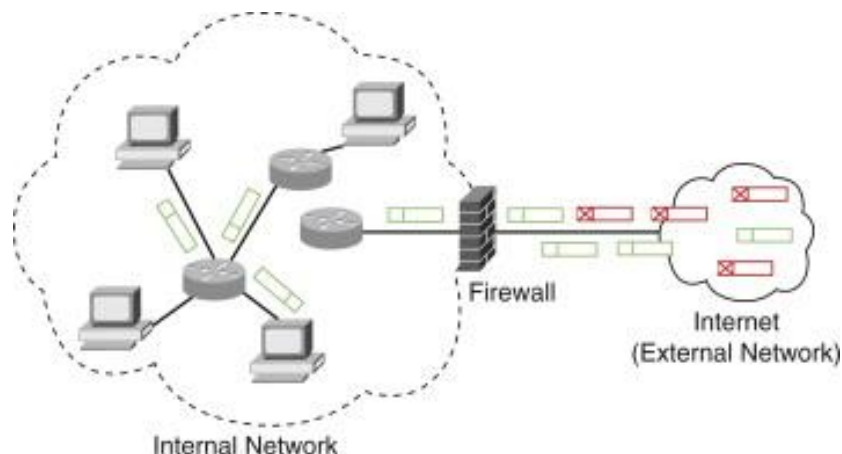
4. Firewall

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and an untrusted external network such as the Internet.

Functions of a Firewall

A firewall can:

- allow or block network traffic
- restrict unauthorised access
- control access to services and resources
- protect systems from external attacks



Common Filtering Rules

Firewalls filter traffic using rules based on:

Filtering Rule	Example
IP address	Allow traffic only from specific computers
Port number	Allow HTTP traffic through port 80
Protocol	Allow HTTPS but block FTP
Payload content	Block suspicious keywords or file types

Types of Firewall

Type	Description
Network-based firewall	Protects an entire network
Host-based firewall	Software installed on an individual computer

Using both network-based and host-based firewalls provides stronger protection.

Limitations of Firewalls

- Cannot fully protect against malware such as viruses or worms
- May not detect insider threats from authorised users
- Encrypted traffic (e.g. HTTPS) can reduce the effectiveness of traffic inspection
- Incorrect firewall rules or misconfiguration may create security gaps
- Firewalls alone cannot provide complete network security and must be combined with other security measures such as antivirus software and IDS/IPS systems



5. Intrusion Detection System (IDS) and Intrusion Prevention System (IPS)

An **Intrusion Detection System (IDS)** monitors network traffic or system activity to detect suspicious or malicious behaviour such as:

- unauthorised access attempts,
- malware activity,
- Denial of Service (DoS) attacks.

When suspicious activity is detected, the IDS sends alerts to network administrators but does not block the attack.

An **Intrusion Prevention System (IPS)** monitors network traffic and automatically blocks malicious activity before it reaches the target system.

An IPS can:

- block malicious packets,
- terminate suspicious connections,
- prevent unauthorised access.

Comparison between IDS and IPS

IDS	IPS
Detects attacks	Detects and blocks attack
Passive monitoring	Active prevention
Sends alerts	Takes immediate action
Attack may still reach target	Attack blocked before reaching target

Types of IDS and IPS

Type	Description
Host-based	Monitors activities on an individual computer or server
Network-based	Monitors traffic across a network

Detection Methods

Method	Description
Signature-based	Detects known attack patterns or signatures
Anomaly-based (Heuristic)	Detects unusual behaviour compared to normal network activity

False Positives and False Negatives

Term	Description	Impact
False positive	Legitimate activity incorrectly identified as malicious	Generates unnecessary alerts
False negative	Actual attack not detected	Allows attacks to go undetected



Limitations and Challenges

IDS and IPS systems have several limitations:

- Encrypted traffic (e.g. HTTPS) may prevent inspection of packet contents.
- Processing and analysing traffic may reduce network performance.
- Attackers may use evasion techniques to bypass detection systems.
- Signature-based systems may fail to detect new or unknown attacks.

6. Encryption, Digital Signature and Authentication

Data transmitted over networks may be intercepted or modified by unauthorised users. Security techniques such as encryption, digital signatures, and authentication help protect data and verify identity.

Encryption

Encryption converts plaintext into ciphertext using an encryption key. The ciphertext can only be read after decryption using the correct key.

Term	Meaning
Plaintext	Original readable message
Ciphertext	Encrypted unreadable message
Encryption	Converting plaintext into ciphertext
Decryption	Converting ciphertext back into plaintext

Symmetric Encryption

Symmetric encryption uses the same key for both encryption and decryption.

- Faster than asymmetric encryption
- Simpler to implement
- Both sender and receiver must know the same key

If the key is compromised, encrypted data can be decrypted.

Asymmetric Encryption

Asymmetric encryption uses a pair of keys:

- a public key,
- a private key.

Data encrypted using one key can only be decrypted using the corresponding key in the pair.

To send encrypted data:

- the sender encrypts the data using the recipient's public key,
- the recipient decrypts it using their private key.



Comparison between Symmetric and Asymmetric Encryption

Symmetric	Asymmetric Encryption
Same key used	Public/private key pair used
Faster	Slower
Simpler	More secure key distribution
Key must be shared securely	Public key can be openly shared

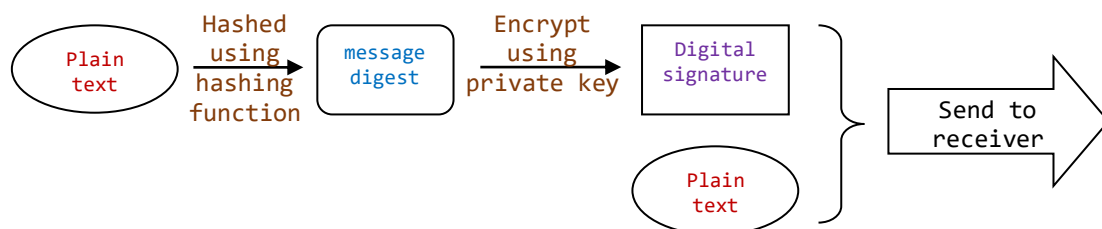
Digital Signature

Digital signatures use asymmetric encryption and hashing to verify:

- authenticity,
- integrity,
- non-repudiation.

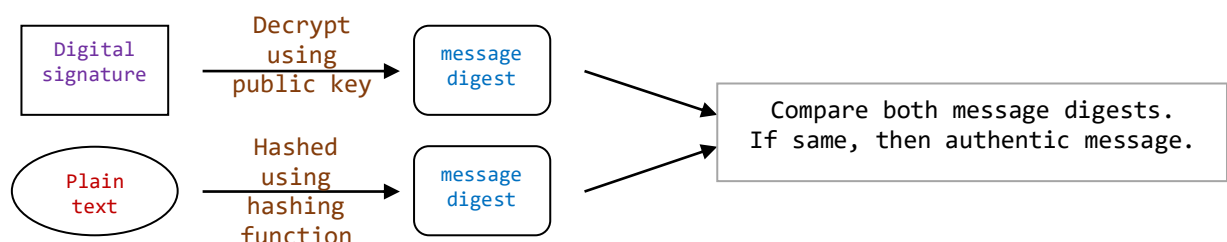
Sender's Side

1. The message is hashed to produce a message digest.
2. The message digest is encrypted using the sender's private key to create the digital signature.
3. The message and digital signature are sent to the recipient.



Recipient's Side

1. The digital signature is decrypted using the sender's public key, resulting in a message digest.
2. The received message is hashed again.
3. The two message digests are compared.
4. If they match, the message is authentic and has not been altered.





Encryption vs Digital Signature

Encryption	Digital Signature
Provides confidentiality	Provides authentication and integrity
Prevents unauthorised reading	Verifies sender identity
Uses recipient's public key	Uses sender's private key

Authentication

Authentication is the process of verifying the identity of a user or system.

Common authentication methods include:

- usernames and passwords,
- one-time passwords (OTP),
- biometric authentication,
- two-factor authentication (2FA).

Authentication helps ensure that only authorised users can access systems and data.

Summary

Security Technique	Main Purpose
Encryption	Confidentiality
Digital Signature	Integrity, authentication, non-repudiation
Authentication	Verify identity

7. Tutorial

1. Define the term malware.
2. State one difference between a virus and a worm.
3. Explain one limitation of a firewall.
4. Explain the purpose of encryption.
5. State one advantage of asymmetric encryption over symmetric encryption.
6. Explain the purpose of a digital signature.
7. State one authentication method other than passwords.
8. Compare symmetric encryption and asymmetric encryption.
9. Compare how a firewall and an intrusion prevention system (IPS) control network traffic.
10. Compare encryption and digital signatures in ensuring network security.
11. A school wants to improve the security of its online learning system. Describe how the following could be used to improve security:
 - firewall,
 - IDS or IPS,
 - encryption,
 - authentication.

For each method, explain one limitation.

[8]



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H2 Computing (9569)

2026 Year 6

12. Customers of a bank have to use the Internet in order to do electronic banking. Explain the problem of maintaining confidentiality of data on the Internet and the need for techniques to address the problem. [5]
13. Explain the difference between an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS). Explain how each system improves network security and state one limitation of each. [6]
14. Explain how firewalls restrict unauthorised access to networks and state two limitations of firewalls. [4]
15. As an employee at a financial institution, you received a loan application from a client electronically.
- (a) Discuss how digital signatures are used to secure and validate the loan application process. [4]
 - (b) Explain the advantages of using digital signatures in this context, both for the financial institution and the client. [4]
16. **9569/01/SP/2020**
- A company has many offices at different sites in many other countries. Each office is able to access data stored on the network servers at the company headquarters. The offices access data using the Internet.
- (a) Describe two threats and how they could compromise company servers. For each threat, explain how the servers and data stored on them could be protected and identify any limitations of the protection. [8]
 - (b) Data transferred over the internet from the company headquarters needs to be kept secure. Explain three security features that should be used in order to ensure this data would be protected before data transfer. [6]
17. Explain why a company should use multiple security measures such as firewalls, encryption, antivirus software, and authentication instead of relying on only one method. [4]
18. Explain how malware or a denial of service (DoS) attack could compromise a computer system or network. State one method that could reduce the risk of each attack. [4]

8. References

- <https://www.cloudflare.com/learning/ddos/what-is-a-ddos-attack/>
- <https://www.zdnet.com/article/what-is-malware-everything-you-need-to-know-about-viruses-trojans-and-malicious-software/>
- <https://www.bankinfosecurity.com/malware-used-to-launch-ddos-attacks-a-8656>